

APPLIED PHYSICS FOR CSE STREAM(BPHYS102/202)

OFFICIAL LINKS

 Website

<https://braintube.site>

 WhatsApp Channel (VTU Notes & Updates)

<https://wa.me/918884624741>

 YouTube Channel

<https://youtube.com/@braintube-v9w?si>

MODULE-3

QUANTUM COMPUTING

BASICS OF QUANTUM INFORMATION & QUBITS

1. Introduction to Quantum Computing

Quantum computing uses principles of **quantum mechanics** to process information.

Unlike classical computers (bits 0 or 1), quantum computers use **qubits**, which can exist in **superposition**.

2. Moore's Law and Its End

Moore's Law states that the number of transistors on a chip doubles approximately every 18–24 months. As transistor sizes approach atomic scales:

- Quantum effects dominate
- Heat dissipation increases
- Further miniaturization becomes impractical

This motivates **quantum computing** as an alternative paradigm.

3. Classical Computing vs Quantum Computing

Classical Computing

- Uses bits (0 or 1)
- Deterministic operations
- Limited parallelism

Quantum Computing

- Uses qubits (superposition of 0 and 1)
- Probabilistic outcomes
- Massive parallelism via superposition and entanglement

4. Concept of a Qubit

A **qubit** is the basic unit of quantum information.

The state of a qubit is written as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where

α, β are complex numbers such that:

$$|\alpha|^2 + |\beta|^2 = 1$$

5. Properties of a Qubit

5.1 Superposition

A qubit can exist in a linear combination of $|0\rangle$ and $|1\rangle$ simultaneously.

5.2 Measurement

Upon measurement:

- Probability of outcome 0 = $|\alpha|^2$
- Probability of outcome 1 = $|\beta|^2$

After measurement, the qubit **collapses** to the measured state.

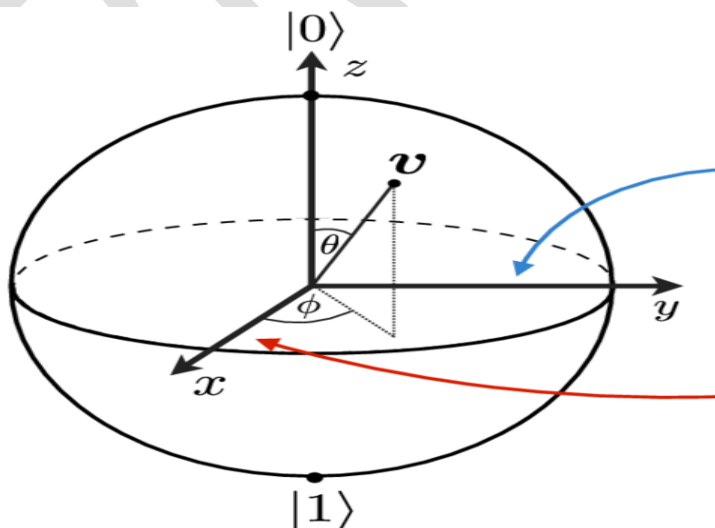
6. Bloch Sphere Representation of a Qubit

Any single qubit state can be represented geometrically on the **Bloch sphere**.

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

- θ and ϕ define the position on the sphere
- North pole $\rightarrow |0\rangle$
- South pole $\rightarrow |1\rangle$

Bloch Sphere Diagram (REQUIRED)



Pole states:

$$\begin{aligned} |i+\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \\ |i-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) \end{aligned}$$

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ |-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \end{aligned}$$

7. Single Qubit States

Computational Basis States

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Any single-qubit state is a superposition of these basis states.

8. Two-Qubit System

The state of a two-qubit system is written as:

$$|\psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle$$

Normalization condition:

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

9. Extension to N Qubits

For an n -qubit system:

- State space dimension = 2^n
- Enables exponential growth in computational capacity

This exponential scaling is the key advantage of quantum computing.

DIRAC REPRESENTATION & MATRIX OPERATIONS

1. Dirac Representation (Ket and Bra Notation)

In quantum mechanics, states are represented using **Dirac notation**.

- **Ket vector:** $|\psi\rangle$
- **Bra vector:** $\langle\psi|$

If:

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

Then:

$$\langle\psi| = [a^* \quad b^*]$$

where $*$ denotes **complex conjugate**.

2. Matrix Representation of $|0\rangle$ and $|1\rangle$ States

The computational basis states are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These form an **orthonormal basis**.

3. Identity Operator I

The **identity operator** leaves a state unchanged.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Operation

$$I|0\rangle = |0\rangle \quad I|1\rangle = |1\rangle$$

4. Pauli Matrices

Pauli matrices are fundamental operators acting on single qubits.

4.1 Pauli-X Matrix

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Operations:

$$X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$$

4.2 Pauli-Y Matrix

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Operations:

$$Y|0\rangle = i|1\rangle \quad Y|1\rangle = -i|0\rangle$$

4.3 Pauli-Z Matrix

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Operations:

$$Z|0\rangle = |0\rangle \quad Z|1\rangle = -|1\rangle$$

5. Conjugate of a Matrix

The **conjugate** of a matrix is obtained by replacing each element with its complex conjugate.

If:

$$A = \begin{bmatrix} 1 & i \\ -2i & 3 \end{bmatrix}$$

Then:

$$A^* = \begin{bmatrix} 1 & -i \\ 2i & 3 \end{bmatrix}$$

6. Transpose of a Matrix

The **transpose** of a matrix is obtained by interchanging rows and columns.

If:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then:

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

7. Unitary Matrix

A matrix U is called **unitary** if:

$$U^\dagger U = I$$

where:

$$U^\dagger = (U^*)^T$$

Unitary matrices preserve **probability** in quantum systems.

8. Row and Column Matrices – Inner Product

Inner Product

The inner product between two states is:

$$\langle \phi | \psi \rangle$$

Example:

$$\langle 0 | 0 \rangle = 1 \langle 0 | 1 \rangle = 0$$

9. Probability in Quantum Mechanics

If a system is in state:

$$| \psi \rangle = \alpha | 0 \rangle + \beta | 1 \rangle$$

Then:

- Probability of measuring $|0\rangle = |\alpha|^2$
- Probability of measuring $|1\rangle = |\beta|^2$

10. Quantum Superposition

A qubit exists in a linear combination of basis states:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

This is called **superposition**.

11. Normalization Rule

For a valid quantum state:

$$|\alpha|^2 + |\beta|^2 = 1$$

This ensures total probability equals unity.

12. Orthogonality

Two states $|\psi_1\rangle$ and $|\psi_2\rangle$ are **orthogonal** if:

$$\langle\psi_1|\psi_2\rangle = 0$$

Example:

$$\langle 0 | 1 \rangle = 0$$

13. Orthonormality

A set of states is **orthonormal** if:

$$\langle\psi_i|\psi_j\rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

The basis states $|0\rangle$ and $|1\rangle$ are orthonormal.

QUANTUM GATES (SINGLE & MULTI-QUBIT)

1. Quantum Gates – Introduction

Quantum gates are **unitary operators** that act on qubits to change their states. They are the quantum analogue of classical logic gates and are **reversible**.

For any quantum gate U :

$$U^\dagger U = I$$

2. Single-Qubit Gates

Single-qubit gates operate on one qubit at a time.

2.1 Quantum NOT Gate (Pauli-X Gate)

Matrix:

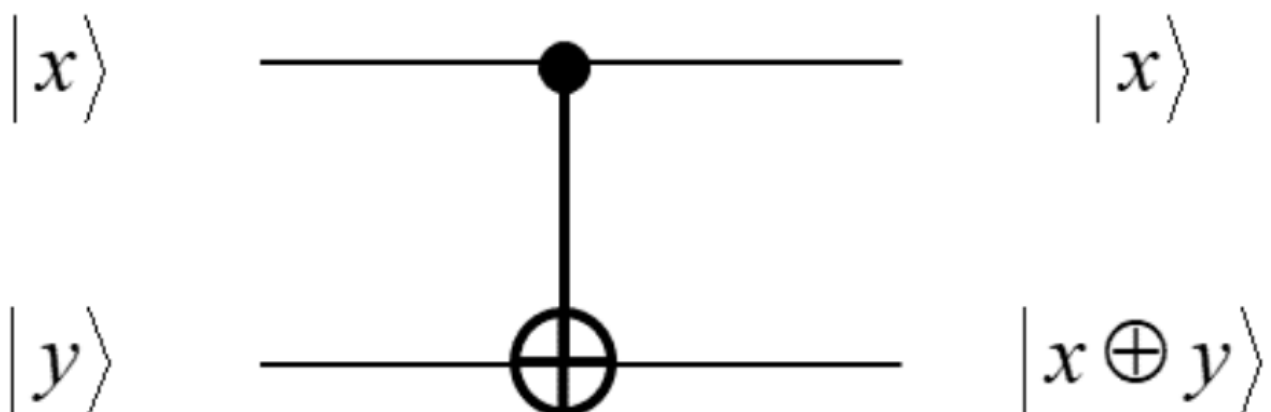
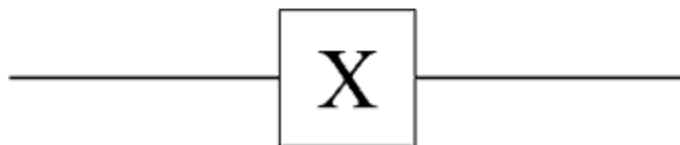
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Action:

$$X | 0 \rangle = | 1 \rangle, X | 1 \rangle = | 0 \rangle$$

This flips the qubit state.

Gate symbol (required):



2.2 Pauli-Y Gate

Matrix:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Action:

$$Y | 0 \rangle = i | 1 \rangle, Y | 1 \rangle = -i | 0 \rangle$$

2.3 Pauli-Z Gate

Matrix:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Action:

$$Z | 0 \rangle = | 0 \rangle, Z | 1 \rangle = -| 1 \rangle$$

2.4 Hadamard Gate (H Gate)

Matrix:

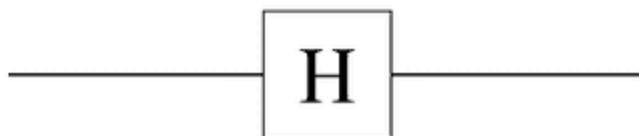
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Action:

$$H | 0 \rangle = \frac{| 0 \rangle + | 1 \rangle}{\sqrt{2}}, H | 1 \rangle = \frac{| 0 \rangle - | 1 \rangle}{\sqrt{2}}$$

Creates **superposition**.

Gate symbol (required):



2.5 Phase Gate (S Gate)

Matrix:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Action:

$$S | 0 \rangle = | 0 \rangle, S | 1 \rangle = i | 1 \rangle$$

2.6 T Gate

Matrix:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Introduces a $\pi/4$ phase shift to $| 1 \rangle$.

3. Multi-Qubit Gates

These gates operate on **two or more qubits** and are essential for **entanglement**.

3.1 Controlled Gate

A controlled gate applies an operation on the **target qubit** only when the **control qubit** is $| 1 \rangle$.

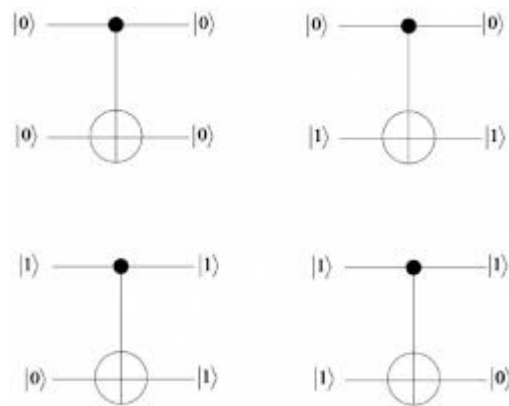
3.2 CNOT Gate (Controlled-NOT)

- Control: first qubit
- Target: second qubit
- Target flips if control = 1

Truth table (four inputs):

Control	Target	Output
0	0	00
0	1	01
1	0	11
1	1	10

CNOT circuit



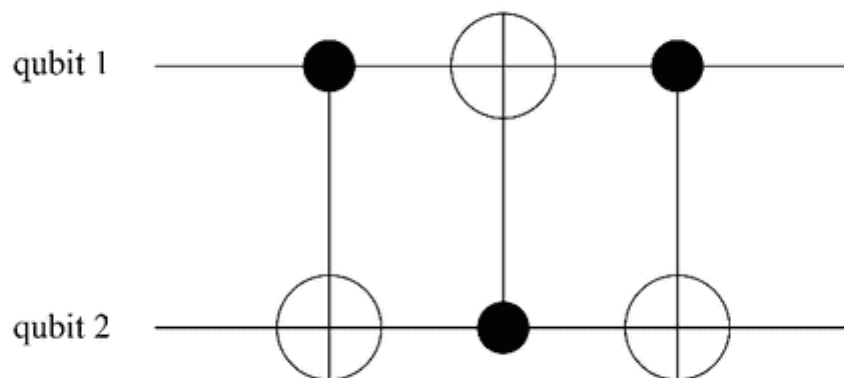
3.3 SWAP Gate

- Exchanges the states of two qubits

Action:

$$|a, b\rangle \rightarrow |b, a\rangle$$

SWAP gate circuit (required):



3.4 Controlled-Z (CZ) Gate

- Applies a phase flip when both qubits are |1⟩

Action:

$$|11\rangle \rightarrow -|11\rangle$$

Other states remain unchanged.

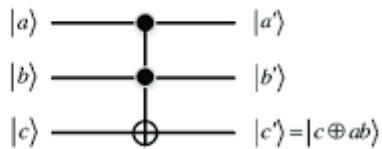
3.5 Toffoli Gate (CCNOT Gate)

- Two control qubits

- One target qubit
- Target flips only if **both controls are 1**

Toffoli circuit (required):

Inputs			Outputs		
a	b	c	a'	b'	c'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0



Quantum circuit diagram for a CCNOT gate. It has three horizontal lines. The first two lines have control dots connected by a vertical line to a target circle on the third line.

$$\text{CCNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4. Importance of Quantum Gates

- Enable superposition and entanglement
- Basis of quantum algorithms
- Building blocks of quantum circuits
- Used in Grover's and Shor's algorithms